

1-Bit Full Adder Design Using Verilog HDL

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Abstract

This project presents the design and simulation of a **1-bit Full Adder** using **Verilog Hardware Description Language (HDL)**. The Full Adder is implemented by cascading two Half Adder modules and an OR gate to generate the final carry output. The design is verified through a behavioral simulation testbench in Xilinx Vivado by applying all possible combinations of the input signals.

Introduction

A Full Adder is a combinational digital circuit that performs the addition of three one-bit binary inputs: **A**, **B**, and **Carry-in (Cin)**. It produces two outputs:

- **Sum**
- **Carry**

Full Adders are fundamental building blocks used in arithmetic logic units (ALUs), processors, digital signal processors, and multi-bit adders.

Objective

- Design a 1-bit Full Adder using Verilog HDL.
 - Implement the Full Adder using two Half Adders.
 - Verify the functionality through simulation.
 - Observe the output waveform and RTL schematic generated in Vivado.
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Design Methodology

The Full Adder is constructed using:

1. **Half Adder 1**

2. Inputs: A and B

3. Outputs:

- Intermediate Sum (S1)
- Carry (C1)

4. **Half Adder 2**

5. Inputs: S1 and Cin

6. Outputs:

- Final Sum
- Carry (C2)

7. **OR Gate**

8. Final Carry = C1 OR C2

Logic Equations

Half Adder

- Sum = $A \oplus B$
- Carry = $A \cdot B$

Full Adder

- Sum = $A \oplus B \oplus \text{Cin}$
 - Carry = $(A \cdot B) + (B \cdot \text{Cin}) + (A \cdot \text{Cin})$
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Truth Table

A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0

A	B	Cin	Sum	Carry
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Verilog Modules

The project consists of three source files:

- `half_adder02.v` – Implements the Half Adder using XOR and AND gates.
- `full_adder02.v` – Builds the Full Adder using two Half Adders and an OR operation.
- `stimulation.v` – Testbench that applies all eight input combinations with a delay of 10 ns between transitions.

Simulation

Behavioral simulation was performed in **Xilinx Vivado**. The testbench sequentially applies every possible input combination and verifies the generated **Sum** and **Carry** outputs.

The simulation waveform confirms that:

- `Sum` follows the XOR of the three inputs.
- `Carry` becomes high whenever at least two inputs are high.

RTL Schematic

The RTL schematic generated in Vivado shows:

- Two instantiated `half_adder02` modules.
- One OR gate combining the carry outputs.
- Correct signal flow from inputs (`A`, `B`, `Cin`) to outputs (`Sum`, `Carry`).

Tools Used

- Verilog HDL
- Xilinx Vivado

- Behavioral Simulation
 - RTL Schematic Viewer
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Results

The implemented Full Adder successfully produces the correct outputs for all input combinations. Simulation waveforms and RTL diagrams validate the correctness of the design.

Applications

- Arithmetic Logic Units (ALUs)
 - Binary Adders
 - Multipliers
 - Digital Processors
 - FPGA and ASIC Design
 - VLSI Digital Systems
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Conclusion

The 1-bit Full Adder was successfully designed and simulated using Verilog HDL. The modular approach using two Half Adders improves readability and reusability while demonstrating key concepts in combinational logic design. The simulation and RTL schematic confirm the functional correctness of the implementation.

GitHub Repository Structure

```
Full-Adder-Verilog/  
├─ half_adder02.v  
├─ full_adder02.v  
├─ stimulation.v  
├─ README.md  
└─ screenshots/  
    ├─ waveform.png  
    └─ rtl_schematic.png
```